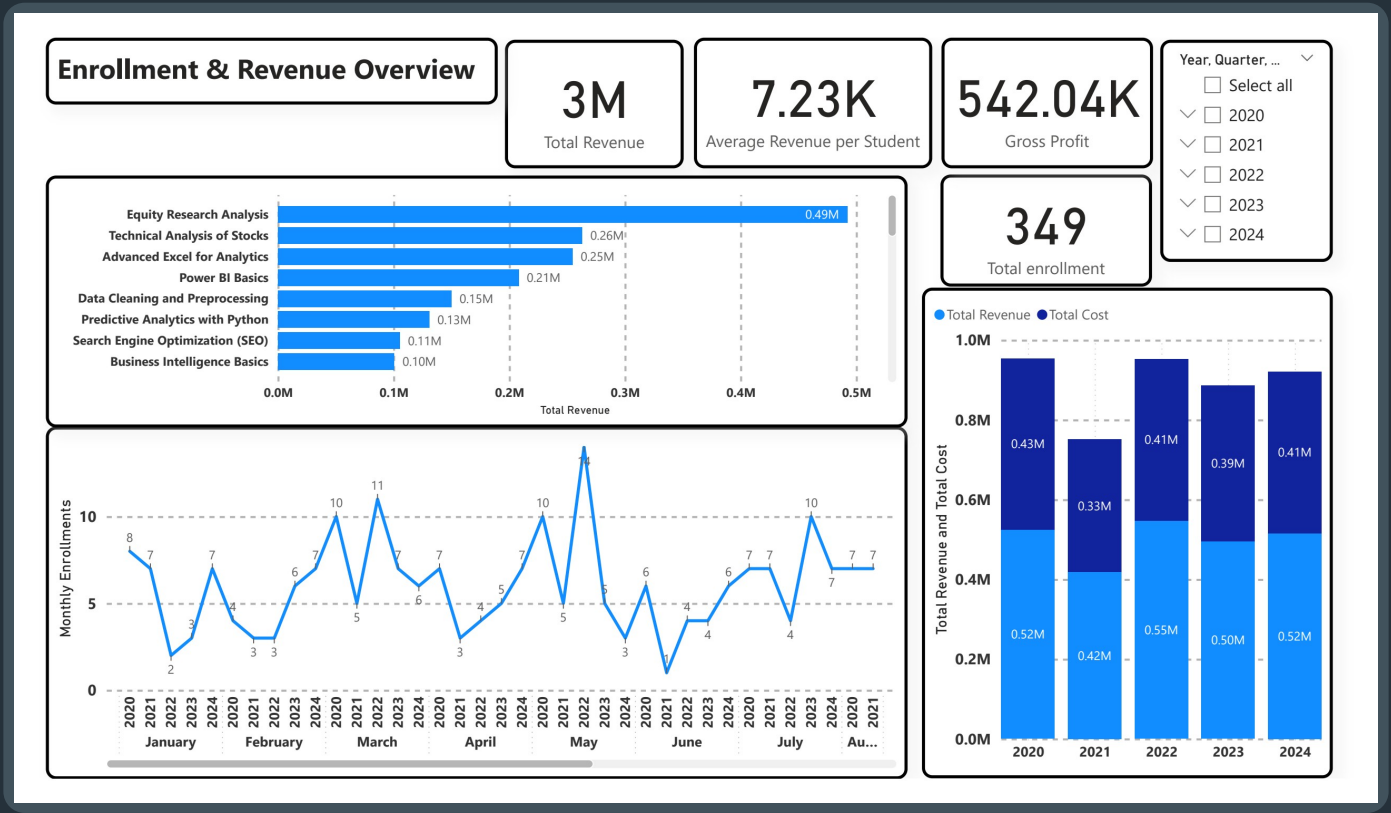
EduSphere Academy
COACHING INSTITUTE · 2020 – 2024

POWER BI PROJECT · END-TO-END DATA ANALYTICS

Institute Performance & ROI Dashboard

Five years of enrollments, courses, trainers and marketing spend turned into an 8-page Power BI dashboard that tells leadership where to invest next.



Microsoft Excel · Power Query · Power BI · DAX · Star schema · Data storytelling

₹2.52M	349	21.5%	36	8
revenue analysed	enrollments	gross margin	courses · 16 trainers	dashboard pages

Pranav Verma

Portfolio project · Power BI

01 · OVERVIEW

Project at a glance

EduSphere Academy (a fictional coaching institute) sells finance, analytics and marketing courses. Its data sat in one wide spreadsheet; leadership wanted a single place to see what's growing, what's profitable and where marketing money works.

₹2.52M
total revenue (fees)

₹542K
gross profit

21.5%
gross margin

98%
course completion rate

349
enrollments · 215 students

36
courses in 6 categories

16
trainers

8
admission sources

Business questions the dashboard answers

- Are enrollments and revenue growing year over year — and is profit keeping pace?
- Which courses and categories earn the most, and which cost more than they bring in?
- Which trainers carry the most students and generate the most profit per student?
- Which admission channels bring volume, and which bring the best return on spend?
- How are students choosing to learn — online, classroom, self-paced or blended?

How I built it

Excel — validate keys, map & de-dupe → **Power Query** — load & shape 13 tables → **Data model** — star schema, 1 fact + 12 dims → **DAX** — 15+ measures & KPIs → **Dashboard** — 8 themed pages with slicers → **Report** — insights & recommendations

Eight pages, one per business area

Page	Focus	Key visuals
1 · Enrollment & Revenue	Enrollment trend, revenue vs cost, top courses	KPI cards, monthly line, stacked columns
2 · Course Performance	Completion, profit and revenue by course	Completion table, category pie, ranked bars
3 · Student Metrics	Completion, gender, certification, duration	Donuts, KPI cards, certification bars
4 · Trainer Utilization	Load, revenue and profit per trainer	Ranked bars, utilization KPI, top-trainer card
5 · Marketing ROI	Cost per acquisition and ROI by channel	CPA bars, source pie, ROI columns
6 · Learning Preferences	Teaching-mode mix and online shift	Donut, stacked areas, % shift KPI
7 · Corporate Training	B2B course revenue and completion	Category donut, fees by course, detail table
8 · Actionable Insights	Red flags: loss-making trainers & courses	Alert KPI, conditional-formatted tables

From one wide sheet to a star schema

The source workbook has a 349-row fact table (29 columns) and 12 lookup sheets. Before modelling, I validated it in Excel so every ID in the fact table has exactly one match.

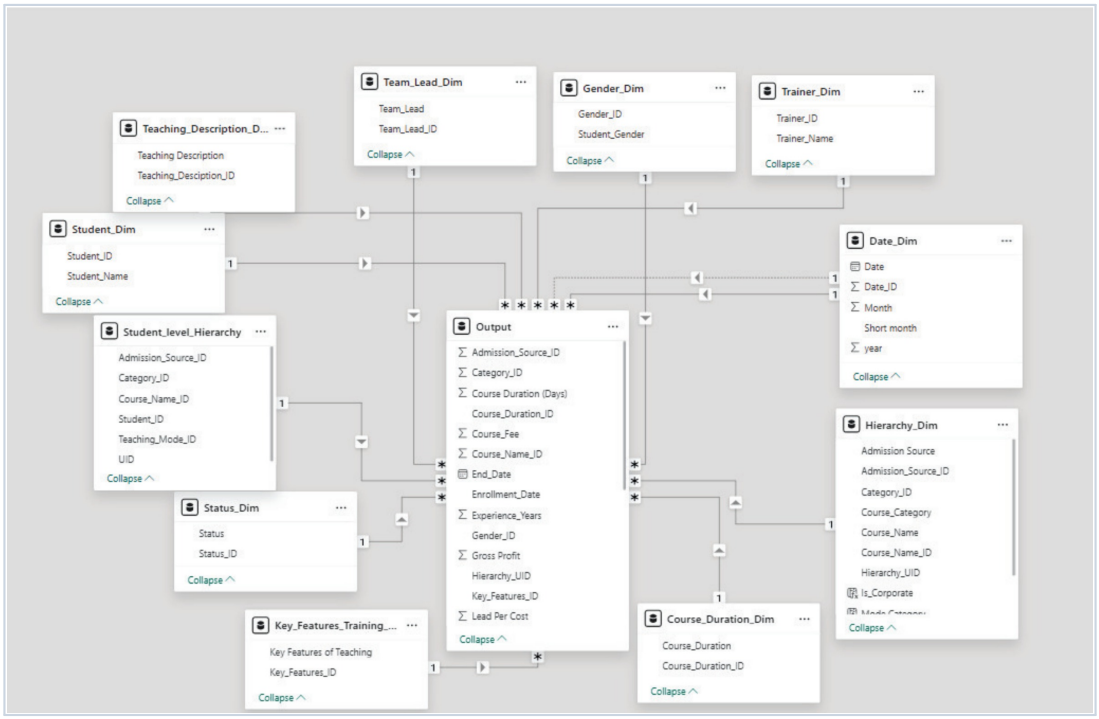
Table	Rows	Contents
Output (fact)	349	One row per enrollment: dates, course, trainer, fees, trainer fee, stationery, lead cost, operating cost, total cost, gross profit, duration
Hierarchy_Dim	52	Course name + category + teaching mode + admission source combinations
Student_level_Hierarchy	350	Maps each student to their course / mode / source combination
Date_Dim	310	Calendar for enrollment, start and end dates
Student · Trainer · Team_Lead	215 · 16 · 9	People dimensions
Gender · Status · Duration · Teaching description · Key features	2 – 5 each	Small lookup tables for slicing and labels

Validation in Excel

- **Student IDs** — every Student_ID in the fact table exists in Student_Dim and the hierarchy table.
- **Dates** — enrollment, start and end dates all map to Date_Dim; invalid dates removed.
- **Foreign keys** — Gender, Course, Status, Trainer and other *_ID columns cross-checked.
- **Composite hierarchy** — Hierarchy_UID mapped to keep course/mode/source segmentation intact.
- **Duplicates** — repeated student-course combinations removed.

Model design

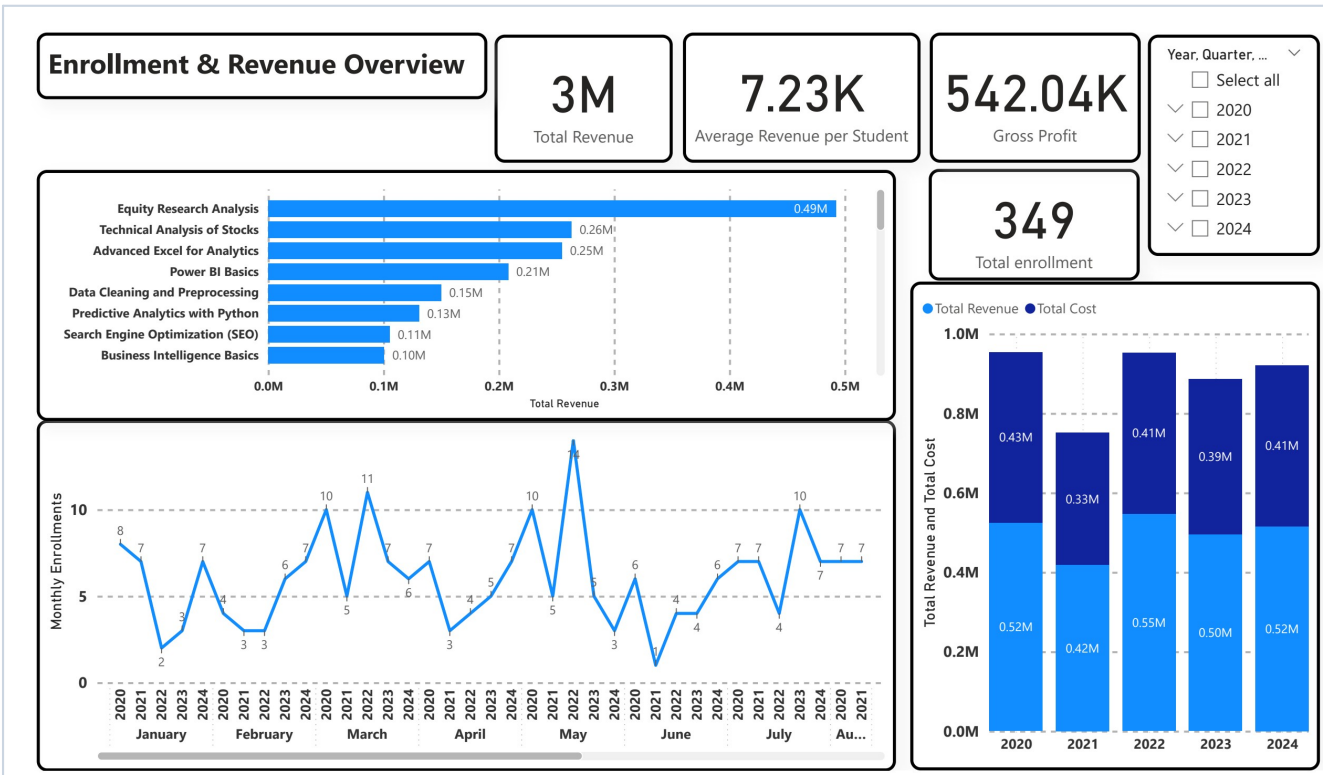
- **Star schema** with `Output` as the single fact table.
- **12 dimensions** joined many-to-one, single direction — predictable filtering and fast DAX.
- Lookup data separated from transactions for a clean, readable model.
- Costs broken out (trainer, stationery, lead, operating) so profit can be explained, not just shown.



Power BI model view: Output fact table in the centre, 12 dimension tables around it

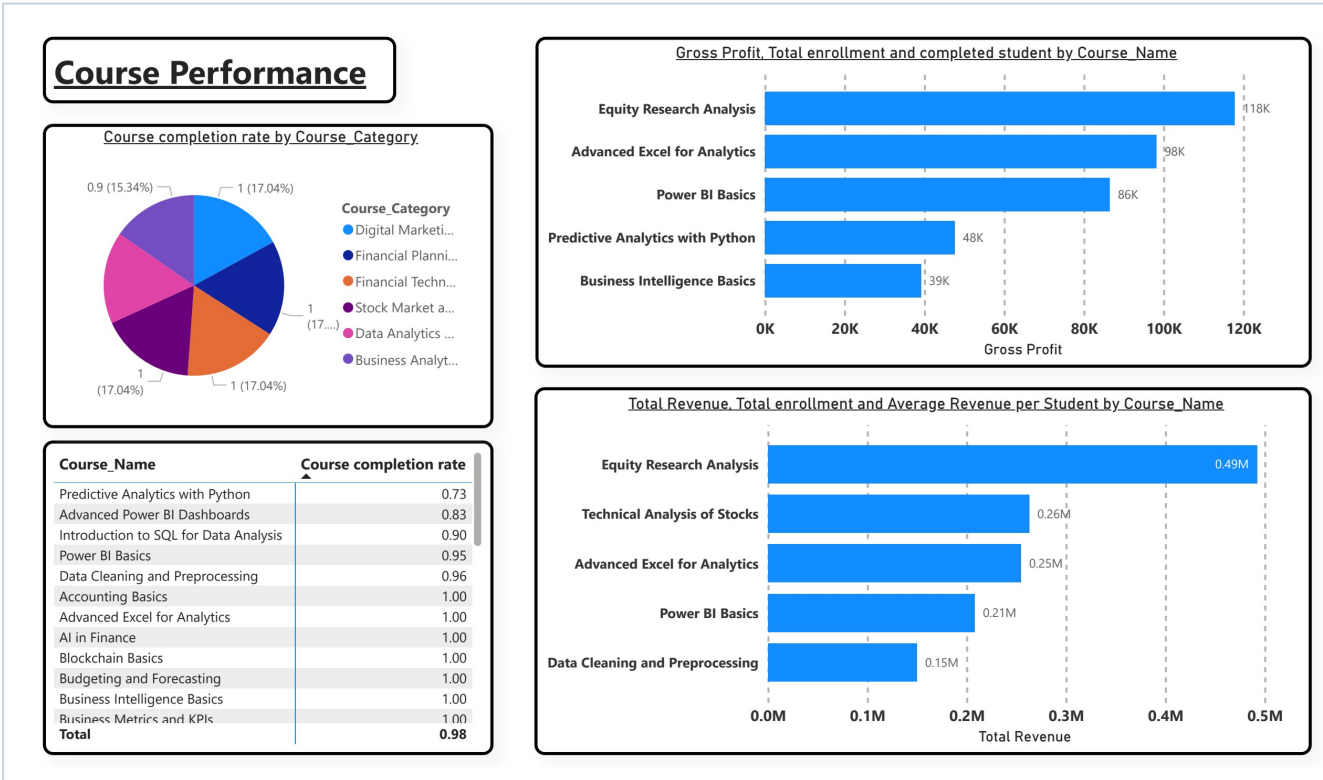
Revenue and course performance

Enrollment & Revenue Overview



KPI cards (revenue, enrollments, revenue per student ₹7.23K, gross profit ₹542K), monthly enrollments across five years, top courses by revenue and revenue vs cost per year. A year/quarter slicer filters every visual.

Course Performance



Completion rate by course and category, gross profit and revenue by course. Equity Research Analysis leads revenue (₹0.49M); Predictive Analytics with Python has the lowest completion rate (73%).

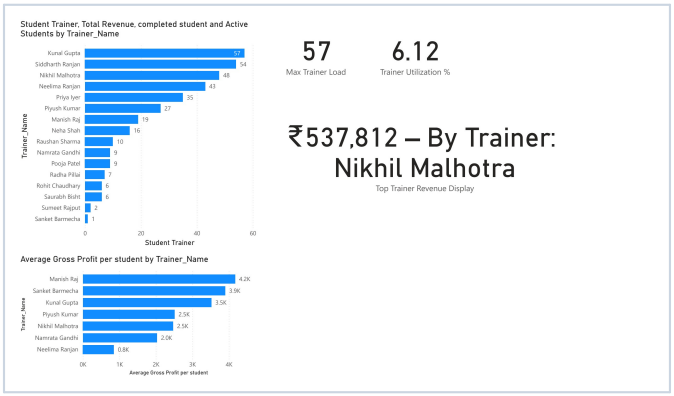
Students, trainers, marketing and learning modes

Student Metrics



98% of students completed; average course length 35.6 days; certification rate by course.

Trainer Utilization



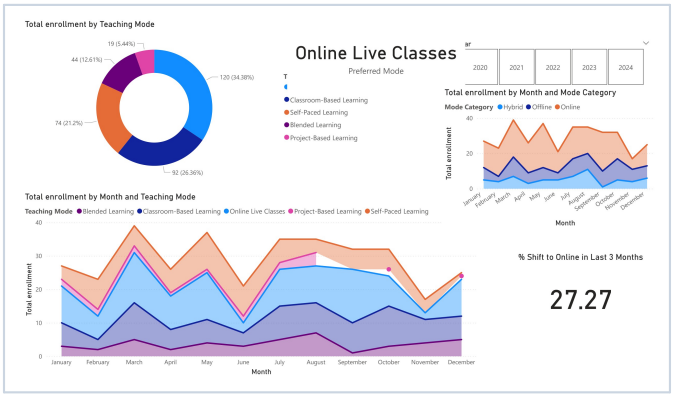
Kunal Gupta trains the most students (57); Nikhil Malhotra brings the most revenue (₹537.8K).

Marketing ROI



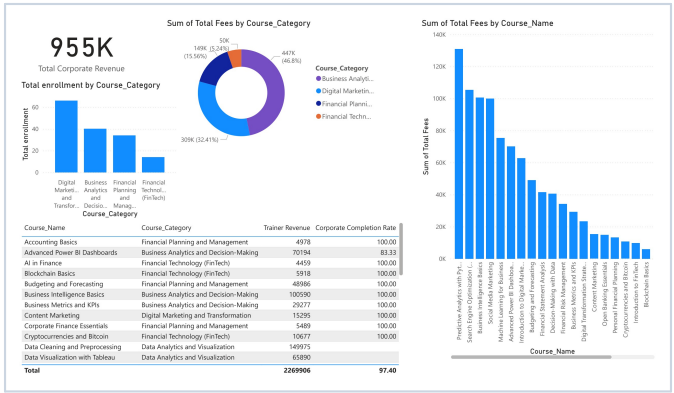
Social Media drives 43% of enrollments; Website Inquiry returns the highest ROI (438%).

Learning Preferences



Online Live Classes is the top mode (34%), with a 27% shift to online in the last 3 months.

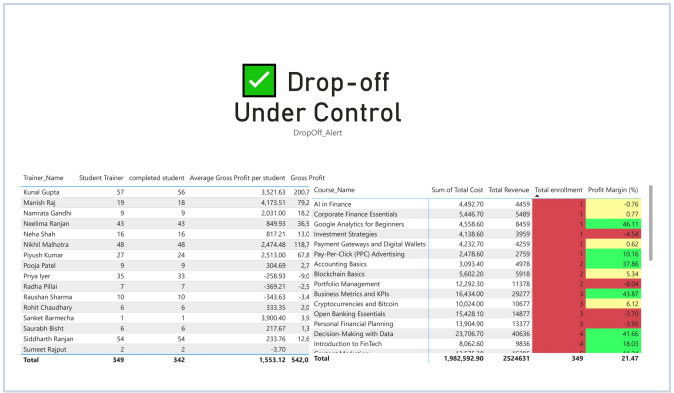
Corporate Training



Course_Name	Course_Category	Trainer Revenue	Corporate Completion Rate
Accounting Basics	Financial Planning and Management	4978	100.00
Advanced Power BI Dashboards	Business Analytics and Decision-Making	70194	83.33
AI in Finance	Financial Technology (FinTech)	4459	100.00
Blockchain Basics	Financial Technology (FinTech)	5918	100.00
Budgeting and Forecasting	Financial Planning and Management	48986	100.00
Business Intelligence Basics	Business Analytics and Decision-Making	100590	100.00
Business Metrics and KPIs	Business Analytics and Decision-Making	29277	100.00
Content Marketing	Digital Marketing and Transformation	15295	100.00
Corporate Finance Essentials	Financial Planning and Management	5489	100.00
Cryptocurrencies and Bitcoin	Financial Technology (FinTech)	10677	100.00
Data Cleaning and Preprocessing	Data Analytics and Visualization	149375	
Data Visualization with Tableau	Data Analytics and Visualization	65890	
Total		**2269906**	**97.40**

₹955K from corporate courses at a 97.4% completion rate, led by Business Analytics.

Actionable Insights



Trainer_Name	Student Trainer	completed student	Average Gross Profit per student	Gross Profit
Kunal Gupta	57	56	3,521.63	200.7
Manish Raj	19	18	4,173.51	79.2
Namrata Gandhi	9	9	2,031.00	18.2
Neelima Ranjan	43	43	849.93	36.5
Neha Shah	16	16	817.21	13.0
Nikhil Malhotra	48	48	2,474.48	118.7
Piyush Kumar	27	24	2,513.00	67.8
Pooja Patel	9	9	304.69	2.7
Priya Iyer	35	33	-258.93	-9.0
Radha Pillai	7	7	-369.21	-2.5
Raushan Sharma	10	10	-343.63	-3.4
Rohit Chaudhary	6	6	333.35	2.0
Sanket Barmecha	1	1	3,900.40	3.9
Saurabh Bisht	6	6	217.67	1.3
Siddharth Ranjan	54	54	233.76	12.6
Sumeet Rajput	2	2	-3.70	
Total	**349**	**342**	**1,553.12**	**542.0**

Course_Name	Sum of Total Cost	Total Revenue	Total enrollment	Profit Margin (%)
AI in Finance	4,492.70	4459	1	-0.76
Corporate Finance Essentials	5,446.70	5489	1	0.77
Google Analytics for Beginners	4,558.60	8459	1	46.11
Investment Strategies	4,138.60	3959	1	-4.54
Payment Gateways and Digital Wallets	4,232.70	4259	1	0.62
Pay-Per-Click (PPC) Advertising	2,478.60	2759	1	10.16
Accounting Basics	3,093.40	4978	1	37.86
Blockchain Basics	5,602.20	5918	1	5.34
Portfolio Management	12,292.30	11378	2	-8.04
Business Metrics and KPIs	16,434.00	29277	2	43.87
Cryptocurrencies and Bitcoin	10,024.00	10677	2	6.12
Open Banking Essentials	15,428.10	14877	3	-3.70
Personal Financial Planning	13,904.90	13377	3	-3.95
Decision-Making with Data	23,706.70	40636	4	41.66
Introduction to FinTech	8,062.60	9836	4	18.03
Total	**1,982,592.90**	**2524631**	**349**	**21.47**

Alert KPI plus conditional formatting that flags loss-making trainers and courses.

04 · DAX

Measures behind the KPIs

Every card and chart is driven by explicit measures rather than implicit column sums, so the same definition of revenue or completion is used on every page.

Measure	Logic	Used for
Total Enrollments	`COUNT(Output[Student_ID])`	Volume KPI and trends
Total Revenue	`SUM of fees in Output`	Revenue KPIs and rankings
Total Cost	`SUM(Output[Total Cost])`	Trainer + stationery + lead + operating cost
Gross Profit	`SUM(Output[Gross Profit])`	Profit KPIs, trainer and course profitability
Avg Revenue per Student	`[Total Revenue] / [Total Enrollments]`	₹7.23K KPI card
Course Completion Rate	`share of rows with Status = "Completed"`	Course and category completion
Drop-off Rate	`1 – [Course Completion Rate]`	Drop-off alert on Actionable Insights
Certification Rate	`completed and certified ÷ enrolled`	Certification by course
Avg Course Duration	`AVERAGE(Output[Course Duration (Days)])`	35.6-day KPI
Cost per Acquisition	`lead cost by admission source`	Marketing efficiency
ROI per Source	`return on lead spend per admission source`	Channel ROI chart
Trainer Utilization	`students ÷ trainers`	Workload KPI
Avg Gross Profit per Student	`AVERAGEX over trainers of [Gross Profit]`	Trainer profitability
Online Share (Last 3M)	`% of enrollments online in the last 3 months`	Online-shift KPI
Top Revenue Trainer	`TOPN(1, SUMMARIZE(…), [Revenue], DESC)`	Dynamic “top trainer” card

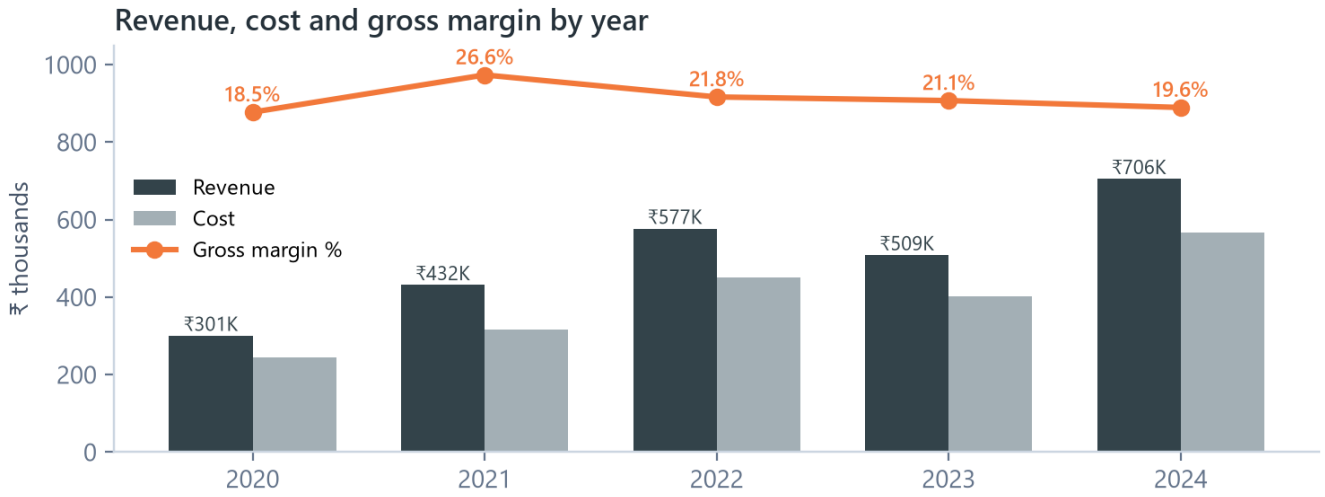
Design principles

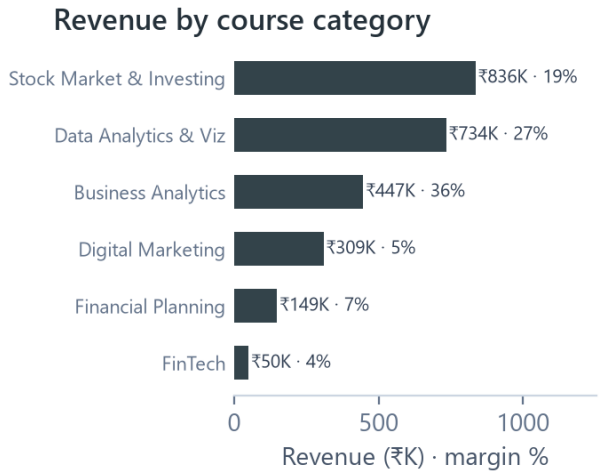
- **One business question per page**, with KPI cards at the top and supporting visuals below.
- **Consistent slicers** — course, status, year and teaching mode filter every page the same way.
- **Tooltips and drill-downs** (year → quarter → month) for detail without clutter.
- **Conditional formatting** turns tables into watch-lists: red for negative margin, green for healthy.

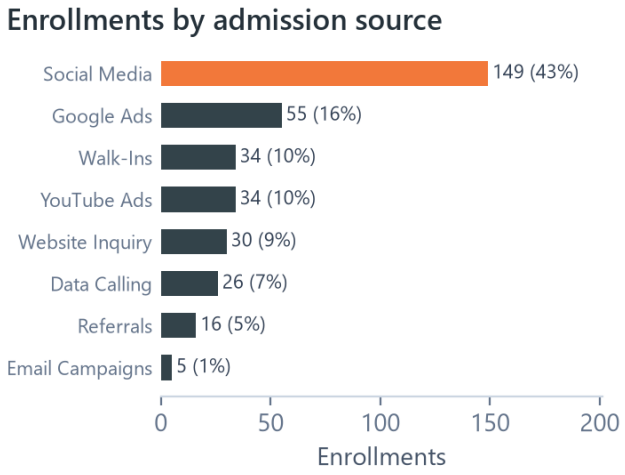
05 · INSIGHTS

What the data says

Recomputed from the source workbook; totals match the dashboard (₹2.52M revenue, ₹542K gross profit, 21.47% margin).







Key takeaways

- **Revenue grew 2.3×** — from ₹301K in 2020 to ₹706K in 2024 — **but margin is slipping**: from a 26.6% peak in 2021 to 19.6% in 2024, because costs are growing faster than fees.
- **Three categories barely break even**: Digital Marketing (5%), Financial Planning (7%) and FinTech (4%) bring in ₹508K of revenue but only about ₹29K of profit.
- **Business Analytics is the most profitable** category at a 36% margin; Data Analytics and Stock Market courses bring the most revenue.
- **Social Media is the volume engine** — 43% of enrollments and ₹1.04M of revenue — while Website Inquiry gives the best ROI (438%).
- **Four trainers run at a loss** (Priya Iyer, Radha Pillai, Raushan Sharma, Sumeet Rajput), together about −₹15K of gross profit.

06 · RECOMMENDATIONS

What I'd tell leadership

01 **Protect the margin while growing**

Revenue doubled but margin fell ~7 points since 2021. Review trainer fees and operating cost per course before the next price or volume push.

02 **Fix or retire low-margin courses**

Redesign or reprice Digital Marketing, Financial Planning and FinTech courses; drop loss-makers like Portfolio Management and Personal Financial Planning.

03 **Shift budget to high-ROI channels**

Keep Social Media for reach, but grow Website Inquiry, Referrals and Email Campaigns, which convert cheaply.

04 **Invest in online delivery**

Online Live Classes is the most chosen mode and the shift to online is accelerating; prioritise online infrastructure and blended formats.

05 **Coach or rebalance under-performing trainers**

Pair the four negative-margin trainers with top performers, or rebalance their course mix toward profitable categories.

06 **Grow corporate training**

₹955K revenue with 97% completion shows strong B2B demand — package Business Analytics and Budgeting courses for companies.

Skills demonstrated

- **Excel** — data validation, key mapping, de-duplication.
- **Power Query** — loading and shaping 13 tables.
- **Data modelling** — star schema, relationships and cardinality.
- **DAX** — KPIs, ratios, AVERAGEX / TOPN patterns.
- **Storytelling** — business-first layout and a written insights report.

What I'd add next

- A **margin-trend** visual so the 2021 → 2024 slide is visible at a glance.
- **Cohort view** of repeat students across courses.
- **Forecast** of next-year enrollments by mode.
- **Drill-through** from a course to its trainers and students.
- Publish to **Power BI Service** with scheduled refresh.